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Assignment - Monte Carlo Simulation to price European Options

Assignment

- Implement Euler's method in Python or MATLAB to price a European call option when the underlying stock follows the dynamics

$$dS_t = rS_t dt + \sigma S_t^\gamma dW_t, \quad (1)$$

The diffusion coefficient, σS_t^γ , models the hypothesis that the volatility depends on the stock price. If $\gamma < 1$, the volatility and the price are inversely related, which has been suggested by empirical evidence. Make use of your results from the Seminar.

You find help on the use of normally distributed random numbers by typing `help(numpy.random.normal)` or `help(random.gauss)` in Python or `help randn` in MATLAB.

As an example you can use the following parameters: $S_0 = 14$, $K = 15$, $r = 0.1$, $\sigma = 0.25$, $T = 0.5$, $\gamma = 0.8$.

- The error in the computed solution has two parts:
 - The sample error* which is due to the fact that you can't use an ∞ number of sample paths.
 - The discretization error/bias* which is due to the fact that you can't use an ∞ number of time-steps.

Use your code to make experiments and measure the error in the computed solution: how the error depends on the number of sample paths and on the time-step? You can use the code `bsexact` (.py or .m) to compute the exact value of the option for $\gamma = 1$. This code uses the exact value of the option $V(s, t)$ for $\gamma = 1$:

$$\begin{aligned} V(s, t) &= sN(d_1) - Ke^{-r(T-t)}N(d_2), \\ d_1 &= \frac{\ln\left(\frac{s}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}, \\ d_2 &= d_1 - \sigma\sqrt{T-t}. \end{aligned}$$

Here $N(x)$ denotes the standard normal cumulative distribution function and is given by $N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{z^2}{2}} dz$.



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- Reduce the error in your computed solution by two methods. (1) Use antithetic variates on (section 6.9.2, p.243 in Hirta or in Assignment 1 in Studium) and to reduce the sample error. (2) Reduce the bias by using another discretization scheme (Milstein or Runge-Kutta in section 6.6).
- Vary γ between 0.5 and 1. What happens to the value of the option?

Additional **tip** on solving the assignment: When checking one type of error convergence, try to make sure that other errors are as small as possible:

- Discretization error can be hard to see, therefore consider a large ($> 10^6$) number of Monte Carlo samples.
- It might also be helpful in these experiments to note that for $\gamma = 1$ you can integrate Geometric Brownian Motion from (1) exactly in time.

You don't have to write a full **report**, but you should hand in your program code and the following plots:

- The sample error as a function of the number of sample paths.
- The discretization error as a function of the time step.
- A plot showing what happens when using antithetic variates.
- A plot showing what happens when using another discretization scheme.
- A plot of V as a function of γ .

All plots should be *commented* with respect to its behavior. How do your obtained results relate to the theory?

Good luck!